

module1_patterns.cpp X

```
1 // 1. square pattern:
#include <iostream>
using namespace std;
int main(){
    int n;
    cout<<"Enter the value of n:";
    cin>>n;
    //outer loop
    for(int i = 0; i<n; i++){
        for(int j = 0; j<n; j++){//inner loop
            cout<<"*";
        }
        cout<<endl;
    }
    return 0;
}
2 //2. HOLLOW SQUARE PATTERN...
32
33 // }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output' & .\module1_patterns.exe

Enter the value of n:5

```
*****
*****
*****
*****
*****
```

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> █

Debug Compile Compile & Run

Ln 1, Col 22 Spa

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```
DSA patterns
module1_patterns.cpp x
C:\code\dsa_patterns\cpp> g++ main.cpp
// 1. square pattern:
//2. HOLLOW SQUARE PATTERN
#include <iostream>
using namespace std;
int main(){
    int n;
    cout<<"enter the value of n:";
    cin>>n;
    for(int i = 0; i<n; i++){
        for(int j = 0; j<n; j++){
            if(i==0 || i==n-1 || j==0 || j==n-1){
                cout<<"*";
            }
            else{
                cout<<" ";
            }
        }
    }
}

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'
PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\module1_patterns.exe
enter the value of n:5
*****
* *
* *
* *
*****
PS C:\Users\dell\Downloads\DSA patterns\Week1\output>

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C++ Go Live
```

module1_patterns.cpp x

Update

```
//      cout<<endl;
//      return 0;
38 // }
//3..TRIANGLE:
#include <iostream>
using namespace std;
int main(){
    int n;
    cout<<"Enter the value of n:";
    cin>>n;
    for(int i =0; i<n; i++){
        for(int j = 0; j<i+1; j++){
            cout<<"*";
        }
        cout<<endl;
    }
    return 0;
}
//A inverted triangle.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\module1_patterns.exe

Enter the value of n:5

```
*
**
***
****
*****
```

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> |

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Go Run ...



1X15 (module1_patterns.cpp)



Update

module1_patterns.cpp x

```
//  
// }  
// return 0;  
// }  
//4.inverted triangle:  
#include <iostream>  
using namespace std;  
int main(){  
    int n;  
    cout<<"enter the value of n:";  
    cin>>n;  
    for(int i =n; i >=0 ; i--){ //outer loop  
        for(int j = 0; j<i; j++){ //inner loop  
            cout<<"*";  
        }  
        cout<<endl;  
    }  
    return 0;  
}  
/* REGULAR INVERTED TRIANGLE:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\module1_patterns.exe

enter the value of n:5

**

*



c:\Users\dell\Downloads\DSA patterns\Week1\output>

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① Compilation successful

Update

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

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Go Run ...

DSA patterns

Update

module1_patterns.cpp x

```
//  
//5.pryamid pattern  
#include <iostream>  
using namespace std;  
int main(){  
    int n;  
    cout<<"enter the value of n:";  
    cin>>n;  
    for(int i = 0; i<n ; i++){  
        for(int j =0; j<n-i-1; j++){ //for spaces  
            cout<<" ";  
        }  
        for(int j = 0; j<2*i+1; j++){  
            cout<<"*";  
        }  
        cout<<endl;  
    }  
    return 0;  
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\module1_patterns.exe

enter the value of n:5

```
*  
***  
*****  
*****  
*****
```

PS C:\Users\dell\Downloads\DSA patterns\Week1\output>

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File Edit Selection View Go Run ... DSA patterns

module1_patterns.cpp

```
//  
// }  
// }  
//6.DIAMOND PATTERN  
#include <iostream>  
using namespace std;  
int main(){  
    int n;  
    cout<<"Enter the value of n:";  
    cin>>n;  
    for(int i = 0; i<n; i++){ // for upper pryamid  
        for(int j = 0; j<n-i-1; j++){  
            cout<<" ";  
        }  
        for(int j= 0; j<2*i+1; j++){  
            cout<<"*";  
        }  
        cout<<endl;  
    }  
    for(int i = n-1; i>0; i--){  
        for(int j = 0; j<=n-i-1; j++){  
            cout<<" ";  
        }  
        for(int j = 0; j<2*i-1; j++){  
            cout<<"*";  
        }  
        cout<<endl;  
    }  
    return 0;  
}
```



```

114 // }
115 //6.DIAMOND PATTERN
116 #include <iostream>
117 using namespace std;
118 int main(){
119     int n;
120     cout<<"Enter the value of n:";
121     cin>>n;
122     for(int i = 0; i<n; i++){ // for upper pryamid
123         for(int j = 0; j<n-i-1; j++){
124             cout<<" ";
125         }
126         for(int j= 0; j<2*i+1; j++){
127             cout<<"*";
128         }
129         cout<<endl;
130     }
131     for(int i = n-1; i>0; i--){

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\'module1_patterns.exe'
Enter the value of n:5

```

*
***
*****
*****
*****
*****
***
*

```

I

File Edit Selection View Go Run ...

DSA patterns

module1_patterns.cpp X

Update

```
//  
//7.half diamond  
#include <iostream>  
using namespace std;  
int main(){  
    int n;  
    cout<<"Enter the value of n:";  
    cin>>n;  
    for(int i = 0; i<n;i++){ // upper half diamond  
        for(int j = 0; j<i+1;j++){  
            cout<<"*";  
        }  
        cout<<endl;  
    }  
    for(int i = n-1; i>0; i--){ // lower half diamond  
        for(int j = i; j>0; j--){  
            cout<<"*";  
        }  
        cout<<endl;  
    }  
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\module1_patterns.exe
Enter the value of n:5

```
*  
**  
***  
****  
*****  
****  
***  
**  
*
```

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> |

Debug Compile > Compile & Run

Ln 154, Col 2 Spaces: 4 UTF-8 CRLF () C++

```

//      }
//8.half inverted diamond
#include <iostream>
using namespace std;
int main(){
    int n;
    cout<<"Enter the value of n:";
    cin>>n;
    for(int i = n; i>0; i--){
        //spaces
        for(int j = 0; j<n-i; j++){
            cout<<" ";
        }
        //print*
        for(int j= 0; j<2*i-1; j++){
            cout<<"*";
        }
        cout<<endl;
    }
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\'module1_patterns.exe'

Enter the value of n:5

**

*

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> █

Compilation successful

Debug Compile Compile & Run

Ln 174, Col 2 Spaces: 4 UTF-8 CRLF {} C++

33°C

module1_patterns.cpp X

```
//  
// NUMBER PATTERN  
//1.SQUARE pattern  
#include <iostream>  
using namespace std;  
int main(){  
    int n;  
    cout<<"enter the value of n:";  
    cin>>n;  
    for( int i = 0; i<n; i++){  
        for(int j= 0; j<n; j++){  
            cout<<"1";  
        }  
        cout<<endl;  
    }  
    return 0;  
}  
//2.INCREMENTING PATTERN:  
// #include <iostream>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\module1_patterns.exe

enter the value of n:5

11111

11111

11111

11111

11111

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> |

module1_patterns.cpp X

```
//  
//2.INCREMENTING PATTERN:  
#include <iostream>  
using namespace std;  
int main(){  
    int n;  
    cout<<"enter the value of n:";  
    cin>>n;  
    for(int i = 0; i<=n; i++){  
        for(int j =1; j<=n; j++){  
            cout<<j <<" ";  
        }  
        cout<<endl;  
    }  
    return 0;  
}  
//3.right triangle:  
// #include <iostream>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\module1_patterns.exe

enter the value of n:5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> |

① Compilation

```

//      }
// }
//3.right triangle:
#include <iostream>
using namespace std;
int main(){
    int n;
    cout<<"Enter the value of n:";
    cin>>n;
    for(int i = 0; i<=n; i++){
        for(int j =1; j<=i; j++){
            cout<<j <<" ";
        }
        cout<<endl;
    }
    return 0;
}
// 4. INVERTED TRIANGLE:
// #include <iostream>
// .....

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\'module1_patterns.exe'

Enter the value of n:5

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

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DSA patterns

module1_patterns.cpp X

```
// , }  
// 4. INVERTED TRIANGLE:  
#include <iostream>  
using namespace std;  
int main(){  
    int start_num = 3;  
    int n;  
    cout<<"enter the value of n:";  
    cin>>n;  
    for(int i = 0; i<=n; i++){  
        for(int j = 0; j<i; j++){  
            cout<<" ";  
        }  
        for(int j = 0; j <n-i; j++){  
            cout<<(start_num + i) <<" ";  
        }  
        cout<<endl;  
    }  
    return 0;  
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'
PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\'module1_patterns.exe'
enter the value of n:5
3 3 3 3 3
4 4 4 4
5 5 5
6 6
7

PS C:\Users\dell\Downloads\DSA patterns\Week1\output>

Compilation su

module1_patterns.cpp

DSA patterns

```
// }  
// }  
//5.sandwich pattern  
#include <iostream>  
using namespace std;  
int main(){  
    int n;  
    cout<<"enter the value of n:";  
    cin>>n;  
    for(int i = 0; i<=2 ; i++){ // upper half  
        for(int j = 0; j<=i; j++){ // print spaces  
            cout<< " ";  
        }  
        for(int j = 0; j<5-(2*i); j++){ //print no.;  
            cout<<(3+i) << " ";  
        }  
        cout<<endl;  
    }  
    // lower half:  
    // i count backward:  
    for(int i = 1; i>=0; i--){  
        for(int j = 0; j<i ; j++){  
            cout<<" ";  
        }  
        for(int j = 0; j <5-(2*i); j++){  
            cout<<3+i <<" ";  
        }  
        cout<<endl;  
    }  
    return 0;  
}  
//6. INCREMENTING PATTERN
```


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module1_patterns.cpp X

module1_patterns.cpp → main()

```
// }
// }
//5.sandwich pattern
#include <iostream>
using namespace std;
int main(){
    int n;
    cout<<"enter the value of n:";
    cin>>n;
    for(int i = 0; i<=2; i++){ // upper half
        for(int j = 0; j<=i; j++){ // print spaces
            cout<<" ";
        }
        for(int j = 0; j<5-(2*i); j++){ //print no.;
            cout<<(3+i) <<" ";
        }
        cout<<endl;
    }
    // lower half:
    // i print backwards.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns> cd 'c:\Users\dell\Downloads\DSA patterns\Week1\output'

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\module1_patterns.exe

enter the value of n:5

3 3 3 3 3

4 4 4

5

4 4 4

3 3 3 3 3

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> |

Compilation success

File Edit Selection View Go Run ...

← DSA patterns

module1_patterns.cpp •

```
// , }  
//6. INCREMENTING PATTERN  
#include <iostream>  
using namespace std;  
int main(){  
    int n;  
    cout<<"enter the value of n:";  
    cin>>n;  
    //upper half order  
    for(int i = 0; i<n ; i++){  
        for( int j= 0; j<=i ; j++){  
            cout<<i+3 <<" ";  
        }  
        cout<<endl;  
    }  
    // lower half  
    // print the i as in mirror as decrement order:  
    for(int i = 3; i>=0; i--){  
        for(int j = 0; j<=i; j++){  
            cout<<i+3 <<" ";  
        }  
        cout<<endl;  
    }  
    return 0;  
}
```

module1_patterns.cpp X

```
// , }  
//6. INCREMENTING PATTERN  
#include <iostream>  
using namespace std;  
int main(){  
    int n;  
    cout<<"enter the value of n:";  
    cin>>n;  
    //upper half order  
    for(int i = 0; i<n ; i++){  
        for( int j= 0; j<=i ; j++){  
            cout<<i+3 <<" ";  
        }  
        cout<<endl;  
    }  
    // lower half  
    // print the i as in mirror as decrement order:  
    for(int i = 3; i>=0; i--){  
        for(int j = 0; j<=i; j++){
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dell\Downloads\DSA patterns\Week1\output> & .\module1_patterns.exe
enter the value of n:5

```
3  
4 4  
5 5 5  
6 6 6 6  
7 7 7 7 7  
6 6 6 6  
5 5 5  
4 4  
3
```

PS C:\Users\dell\Downloads\DSA patterns\Week1\output>

2. Star Patterns →

iii) Square pattern:

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Code →

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter the value of n:";
    cin >> n;
    //outer loop
    for (int i=0; i<n; i++) {
        //inner loop
        for (int j=0; j<n; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

Logic → Outer loop (i) → controls the current row.
Inner loop (j) → controls the items inside that row.

For n=5
10 Row 1 (i=0)
when i=0,

j will be initialize at 0 and runs while $j < 5$. (values 0, 1, 2, 3, 4).

- It prints * * * * * (5 asterisks).
- Inner loop finishes.
- `cout << endl;` executes, moving to next line.

2. Rows 2, 3, 4, 5 ($i=2, 3, 4$)

- Exact process repeats.
- Each iteration creates an identical row of 5 stars.

3. When $i=5$

- $5 < 5$ (False).
- loop stop, & the program terminates.

Ques] HOLLOW SQUARE PATTERN ->

Code ->

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
    int n;
```

```
    cout << "enter the value of n:";
    cin >> n;
```

```
    for (int i = 0; i < n; i++) { // outer loop
        for (int j = 0; j < n; j++) { // inner loop
```

```
            if (i == 0 || i == n - 1 || j == 0 || j == n - 1) {
```

```
                cout << "*";
```

```
            }
```

```
            else {
```

```
                cout << " ";
```

```
            }
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

```
* * * * *
*
*
*
* * * * *
```

Logic -> Row 1 -> i = 0 [For n = 5]
for column -> j = 0, 1, 2, 3, 4
hence i == 0 is True

Output *****

Row 2 -> i = 1 j = 0 j == 0 is True
o/p *

Row 3 -> i = 1 j = 1, 2, 3 All False
o/p spaces

Row 2: when $i=1$ $j=4$ ($j=(n-1)$) True
o/p *

For Row 3/4 same rule as R2

For Row 5: $i=4$ $j=(n-1)$ True
 $j=0, 1, 2, 3, 4$
o/p * * * * *

Triangle pattern

Code:-

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter the value of n:";
    cin >> n;
    // outer loop
    // inner loop
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < i + 1; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

Logic:- For $n=5$

Row 1 ($i=0$): Inner loop runs $j=0$.
print *

Row 2 ($i=1$): Inner loop runs $j=0, 1$.
print * *

Row 3 ($i=2$): Inner loop runs $j=0, 1, 2$.
print * * *

Row 4 & 5

$i=3$
 $i=4$

* * * *

* * * * *

$j=0, 1, 2, 3$

$j=0, 1, 2, 3, 4$

(iv) Inverted triangle →

```
* * * * *
* * * *
* * *
* *
*
```

Code :-

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter the value of n:";
    cin >> n;
    //outer loop
    for (int i = n; i >= 0; i--) {
        //inner loop
        for (int j = 0; j < i; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

Logic :-

For n=5

Row 1 When i=5 j loops from 0 to 4 →
Print 5 stars.

Row 2 When i=4 j loop from 0 to 3
Print 4 stars

For Row 3/4/5 same procedures.

Regular Inverted triangle

Code:

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter the value of n:";
    cin >> n;
    // Outer loop for (int i = n; i >= 0; i--) {
    // for spaces for (int j = 0; j < n - i; j++) {
        cout << " ";
    }
    // For print * for (int k = 0; k < i; k++) {
        cout << "* ";
    }
    cout << endl;
}
return 0;
```

```
* * * * *
 * * * *
  * * *
   * *
    *
```


(10) PYRAMID PATTERN →

```
      *
     ***
    *****
   ********
  **********
```

Code →

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter the value of n:";
    cin >> n;
    // outer loop
    for (int i = 0; i < n; i++) {
        // For spaces
        for (int j = 0; j < n - i - 1; j++) {
            cout << " ";
        }
        // For "*"
        for (int j = 0; j < 2 * i + 1; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

Logic → (i) Row (i) from 0 to $n-1$: For $n=5$

(ii) loop for (int $j=0$; $j < n-i-1$; $j++$) prints a decreasing number of spaces to shift the stars right.

(iii) Print stars : $j=0$ to $2*i+1$ print odd no's of '*'. For $n=5$

DIAMOND PATTERN ->

Code:-

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter value of n:";
    cin >> n;

    // Upper Diamond
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            cout << " ";
        }
        for (int j = 0; j < 2 * i + 1; j++) {
            cout << "*";
        }
        cout << endl;
    }

    // Lower Diamond
    for (int i = n - 1; i > 0; i--) {
        for (int j = 0; j <= n - i - 1; j++) {
            cout << " ";
        }
        for (int j = 0; j < 2 * i + 1; j++) {
            cout << "*";
        }
        cout << endl;
    }

    return 0;
}
```

```

      *
     ***
    *****
   *********
  **********
 * *********
*  *****
*   ***
*    *

```

Page No. _____
Date: ____/____/____

Logic for upper diamond: (For $n=5$)

- Outer loop (i from 0 to $(n-1)$): Control Row flow.
- Space loop (j from 0 to $(n-i-1)$): print a decreasing number of spaces.
- Star loop (j from 0 to $2*i+1$): print an increasing, odd number of stars.

Row 0 ($i=0$) $\rightarrow 2(0)+1 = 1$ star

Row 4 ($i=4$) $\rightarrow 2(4)+1 = 9$ stars

Lower diamond:

- Starts from $n-1$ and counts backward down to 1. Reverses the whole progression.
- Space loop $\div$ j from 0 to $n-i-1$: print inc. no. of spaces as it goes down.
- Star loop $\div$ j from 0 to $2*i-1$: prints a decreasing, odd number of stars.

iii) HALF DIAMOND →

Code →

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter value: ";
    cin >> n;

    // upper half
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < i + 1; j++) {
            cout << "*";
        }
        cout << endl;
    }

    // lower half
    for (int i = n - 1; i > 0; i--) {
        for (int j = i; j > 0; j--) {
            cout << "*";
        }
        cout << endl;
    }

    return 0;
}
```

```
*
* *
* * *
* * * *
* * * * *
* * * * *
* * * *
* * *
* *
*
```

[For n=5]

Logic → upper half

// outer loop runs n times

// inner loop Row 0 (i=0) → print 0+1 = 1 star

Row (n-1) → print (n-1)+1 = n stars

Lower half

// outer loop $\div$ Counts down from $n-1$ to 1.

// inner loop $\rightarrow$

First lower row ($i = n-1$) $\rightarrow$ print $(n-1)$ stars

last row ($i = 1$) $\rightarrow$ print 1 star.

(viii) HALF INVERTED DIAMOND →

```
* * * * *
 * * * *
  * * *
   * *
    *
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter the value of n:";
```

```
    cin >> n;
```

```
    // outer loop
```

```
    for (int i = n; i > 0; i--) {
```

```
        // spaces
```

```
        for (int j = 0; j < n - i; j++) {
```

```
            cout << " ";
```

```
        }
```

```
        // print "*"
```

```
        for (int j = 0; j < 2 * i + 1; j++) {
```

```
            cout << "*";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

logic $\rightarrow$ $\text{Fou}(n=5)$
// outer loop runs from n to 1

// space loop

Row 1 : Print $\Rightarrow n - n = 0$ space

Row 2 : Print $\Rightarrow n - (n-1) = 1$ space

Row 5 : Print $\Rightarrow n - (n-4) = 4$ space

// Print $\star$

Start from 0 to $2*n-1$

Row 1 ($i=5$) : $2*n-1 = 9$ star.

Last row ($i=1$) : $2*n-1 = 1$ star.

NUMBER PATTERN:->

(A) SQUARE pattern:->

Code:->

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter value of n:";
    cin >> n;
    //outer loop
    //inner loop
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            cout << "1";
        }
        cout << endl;
    }
    return 0;
}
```

```
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
```

Logic:-

Row 1:- for (i=0) [For n=5]
j runs 0 to (n-1)
print 1 1 1 1 1

for (i=1)
j runs 0 to (n-1)
1 1 1 1 1

1

for (i=4)
j runs 0 to (n-1)
1 1 1 1 1

12) INCREMENTING PATTERN-2

Code ->

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "enter value of n:";
    cin >> n;
    //outer loop
    for (int i = 0; i <= n; i++) {
        //inner loop
        for (int j = 1; j <= n; j++) {
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}
```

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

logic ->

for Row 1 ($i=0$)

j runs 1 to (n)

print j

// 1 2 3 4 5

Again Row 2 ($i=1$)

j runs 1 to (n)

print j

// 1 2 3 4 5

last Row ($i=4$)

j runs 1 to (n)

print j

// 1 2 3 4 5

iii) Right triangle:

Code →

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter value of n:";
    cin >> n;
    // outer loop
    for (int i = 0; i <= n; i++) {
        // inner loop
        for (int j = 1; j <= i; j++) {
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Logic :- Row 1 ($i=0$) to n

j runs from 1 to $\leq i$
print 1

Row 2 ($i=1$)

j runs from 1 to $k=i$
1, 2

last row ($i=4$)

j runs from 1 to $\leq i$
1 2 3 4 5

For $n=5$

(iv) INVERTED TRIANGLE :-

3 3 3 3 3

4 4 4 4

5 5 5

6 6

7

Code :-

```
#include <iostream>
using namespace std;
int main() {
    int start_num = 3;
    int n;
    cout << "Enter value of n: ";
    cin >> n;
    // outer loop
    // spaces
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < i; j++) {
            cout << " ";
        }
        // print No.
        for (int j = 0; j < n - i; j++) {
            cout << (start_num + i) << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Logic :-

For n=5

- Outer loop runs from 0 to n
- Now print spaces in inc. order.
- Print No.

Row 1 ($i=0$)

// spaces

// NO.s

$j=0$ print 0 spaces
and $j=0$ to $n-i$ ($j++$)
 $3+i$

// 3 3 3 3 3

cout << endl;

Row 2 ($i=1$)

// spaces

// NO.s

$j=1$ print 1 space
 $j=0$ to $n-i$
 $3+i$

// 4 4 4 4

last Row ($i=4$)

// spaces

// NO.s

$j=0$ to $n-i$
print 4 spaces
 $3+i$

// 7

lv) Sandwich pattern -

3 3 3 3 3

4 4 4

5

4 4 4

3 3 3 3 3

Code:-

```
#include <iostream>
using namespace std;
int main()
```

```
{
    int n;
```

```
    cout << "Enter the value of n:";
```

```
    cin >> n;
```

```
    // Upper half:-
```

```
    // Spaces
```

```
    for (int i = 0; i <= 2; i++) {
```

```
        for (int j = 0; j <= i; j++) {
            cout << " ";
```

```
        }
```

```
    // Print NO
```

```
        for (int j = 0; j < 5 - (2 * i); j++) {
            cout << (3 + i) << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    // Lower half:-
```

```
    // i count backward:-
```

```
    // Spaces
```

```
    for (int i = 1; i >= 0; i--) {
```

```
        for (int j = 0; j < i; j++) {
            cout << " ";
```

```
        }
```

```
    // Print NO
```

```
        for (int j = 0; j < 5 - (2 * i); j++) {
            cout << 3 + i << " ";
```

```
        }
```

```
        cout << endl;
```


} return 0;

Logic :- For $n=5$

// upper half $\rightarrow$ i runs from 0 to 2 [outer loop]

// spaces $j=0$ to $j \leq i$
// No's $j=0$ to $5-(2*i)$

Row 1 $\rightarrow i=0$

// spaces j runs from 0 print 0 spaces
// No's j runs from 0 to 5 // 3 3 3 3 3
print($3+i$)

// lower half $\rightarrow$

i from 1 to 0 "backward"

// spaces for $i=1$

j runs 1 time - 1 space

// No's $j=0$ to $5-(2*i)$
 $< 3+i$; // 4 4 4

(vi) INCREMENTING PATTERN

Code :-

```
#include <iostream>
using namespace std;
int main() {
```

```
    int n;
```

```
    cout << "Enter value of n:";
```

```
    cin >> n;
```

```
    // upper half order
```

```
    for (int i = 0; i < n; i++) {
```

```
        for (int j = 0; j <= i; j++) {
```

```
            cout << i+1 << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    // lower half
```

```
    // print i as in mirror
```

```
    for (int i = 3; i >= 0; i--) {
```

```
        for (int j = 0; j <= i; j++) {
```

```
            cout << i+1 << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
}
```

```
    return 0;
```

3

4 4

5 5 5

6 6 6 6

7 7 7 7 7

6 6 6 6

5 5 5

4 4

3

Logic - Upper half -

// outer loop runs from 0 to n
// inner loop j runs from 0 to $\leq i$

Row 1 $\Rightarrow (i=0)$

j runs 0 to i
print cout $\ll i+3$
// 3

!
last row $(i=4)$

j runs 0 to i
// 7 7 7 7 7

// lower half -

i in backward trace

// outer loop runs from 3 to 0
// inner loop 0 to $\leq i$

R: 3 $(i=3) \Rightarrow$ j runs from 0 to $\leq i$
cout $\ll i+3$

// 6 6 6 6

!
last row $(i=0) \Rightarrow$ j runs from 0 to $\leq i$
cout $\ll i+3$ // 3